

ClusterBrain - Architectural & Design Choices

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1. System Architecture: FastAPI + Streamlit

We explicitly chose to strictly **decouple the system** into a stateful backend and a stateless frontend.

- **FastAPI (Backend):** Selected for its robust capability to handle high-concurrency state simulations and asynchronous task routing without lag. All critical state management, algorithmic scheduling, and queue evaluations happen entirely in-memory on the backend, making it highly resilient.
- **Streamlit (Frontend):** Selected to enable real-time, stateful telemetry visualization. By keeping the frontend completely stateless, we avoided the overhead of complex frontend state management (e.g., Redux). The dashboard simply polls the FastAPI engine and dynamically renders the latest cluster health and operations.

2. Algorithmic Design: Energy-Aware Bin Packing

Rather than using a standard round-robin or least-loaded scheduler, we engineered a custom "**Smart Scheduler**" based on a **4-pass sequential filling algorithm**.

- **The Design Choice:** The scheduler packs jobs sequentially into main nodes until they hit a **70-80% utilization ceiling** before it even considers waking up reserve nodes.
- **The 'Why':** Distributing workloads evenly across all servers keeps the entire cluster awake, burning maximum baseline power. By densely packing a few active nodes, this algorithm mathematically minimizes active power draw and allows the standby nodes to remain completely asleep, maximizing energy savings and reducing carbon footprint.

3. AI Logic: Thermal Triggers vs. Utilization

For our **TabPFN Predictive Operations**, we deliberately programmed the AI to evaluate anomalies strictly based on thermal limits.

- **The Design Choice:** Auto-migration and failure countdowns are triggered **ONLY by critical thermal limits ($\geq 80^{\circ}\text{C}$)**, completely decoupling the anomaly detection from high GPU utilization metrics.
- **The 'Why':** In real-world data centers, ML and deep learning workloads naturally sustain 90%+ GPU utilization for hours or days. Triggering alarms based on high load would result in continuous **false positives**. By isolating the thermal threshold as the ground truth for hardware degradation, the AI only intervenes when a physical hardware meltdown is truly imminent.

4. The 'Right Size Proof' Module Design

To tackle the challenge of proving actual resource savings, we designed a passive tracking mechanism rather than an active blocking routine.

- **The Design Choice:** We implemented an interception hook inside the **"Lazy Evaluation"** job expiration phase. When a job expires, the system simulates realistic actual usage, mathematically caps it so it is always **$\leq$ requested**, and records the data before clearing the node.
- **The 'Why':** This non-blocking approach ensures the core Smart Scheduler and auto-queue are never slowed down. It allowed us to safely verify **Advice Accuracy (OOM safety)** and calculate exact, verified dollar savings with zero performance overhead to the main cluster simulation.